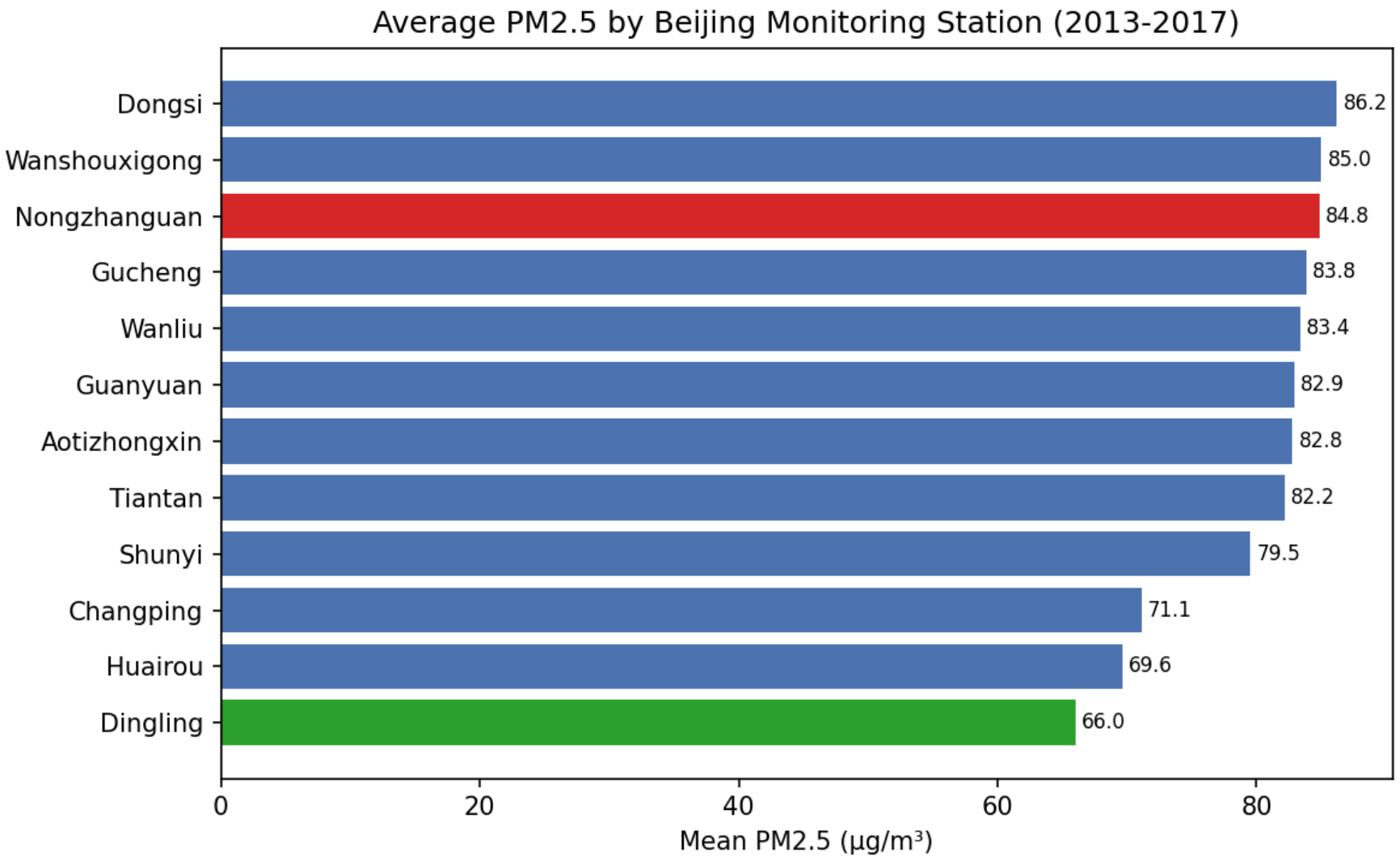
# Which station has the worst air quality?

Dataset: PRSA — 12 monitoring stations across Beijing
Period: 1 Mar 2013 – 28 Feb 2017 (hourly observations)
Pollutants analyzed: PM2.5, PM10, $SO_2$, $NO_2$, CO, $O_3$
Method: per-station mean of each pollutant + a 0–1 normalized composite pollution index

**Nongzhanguan is the worst; Dingling is the cleanest.**

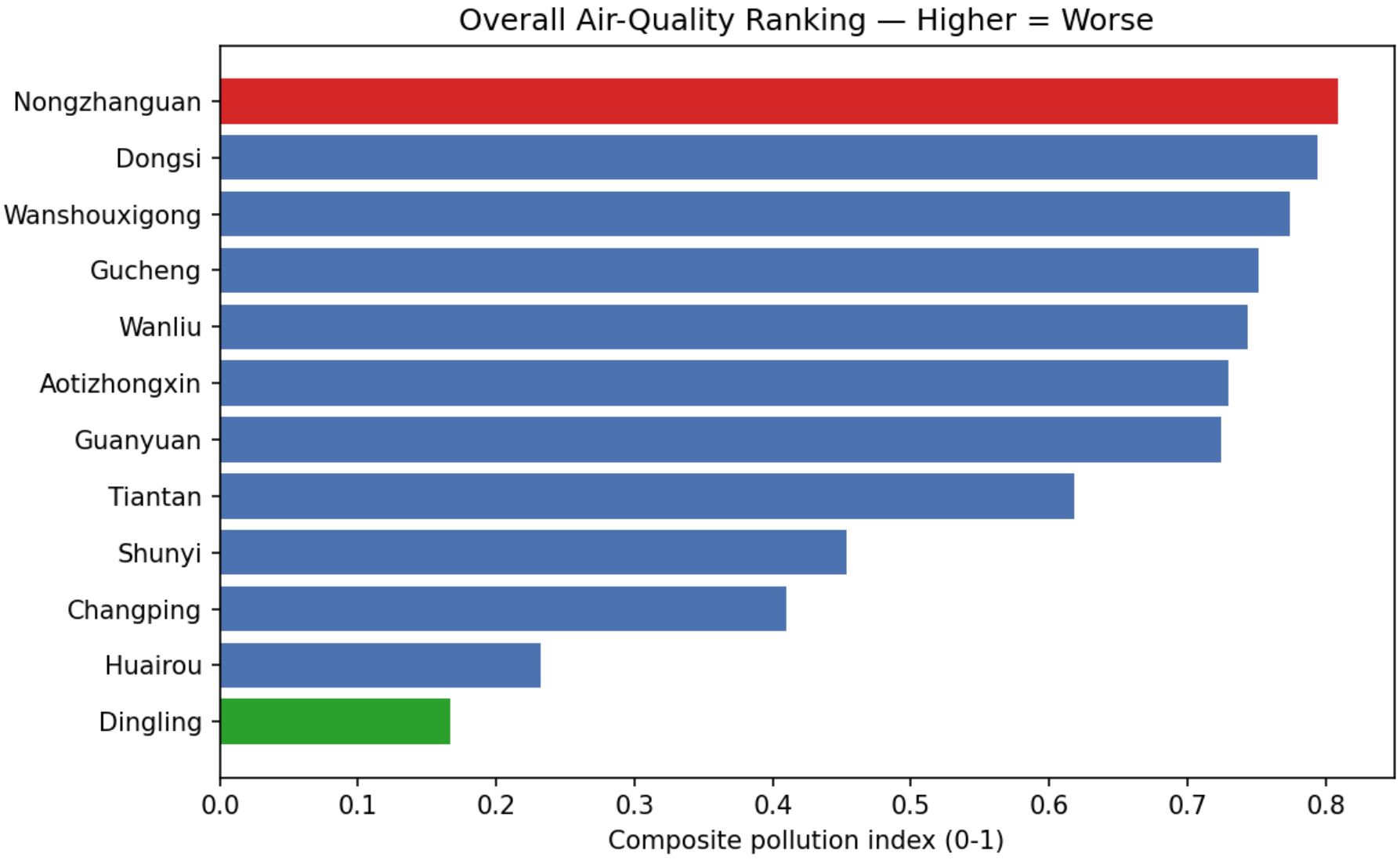
# PM2.5 Ranking by Station




**Key points**

- Urban core stations **(Dongsi, Nongzhanguan, Wanshouxigong, Gucheng)** all exceed 83 µg/m³.
- Suburban/rural sites **(Dingling, Huairou, Changping)** stay below 72 µg/m³.
- WHO annual guideline is 5 µg/m³ — every station is more than 13× over.

# Worst Station: Nongzhanguan




**Nongzhanguan tops the composite index (0.81); its $NO_2$, $SO_2$ and CO are all above the Beijing average.**

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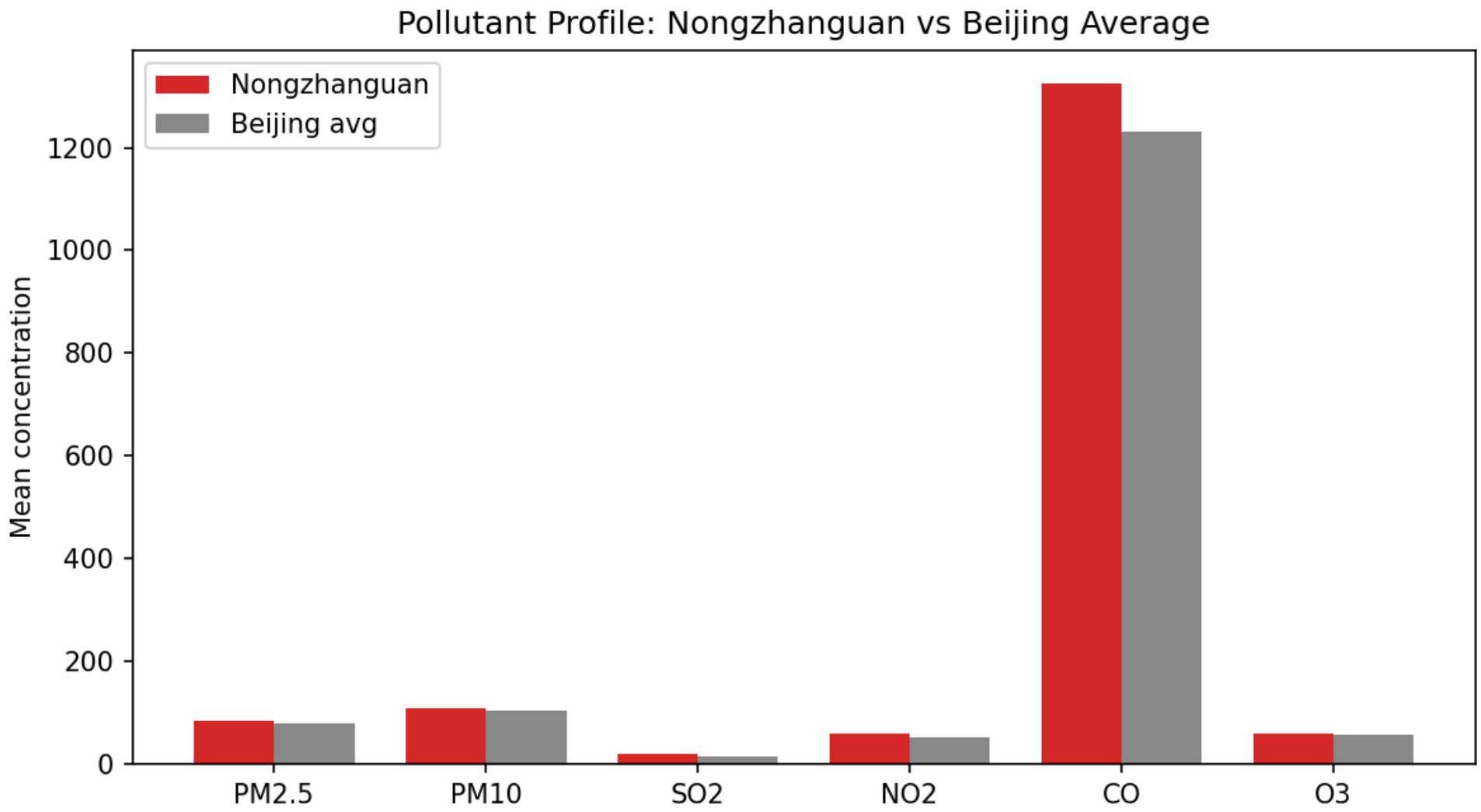


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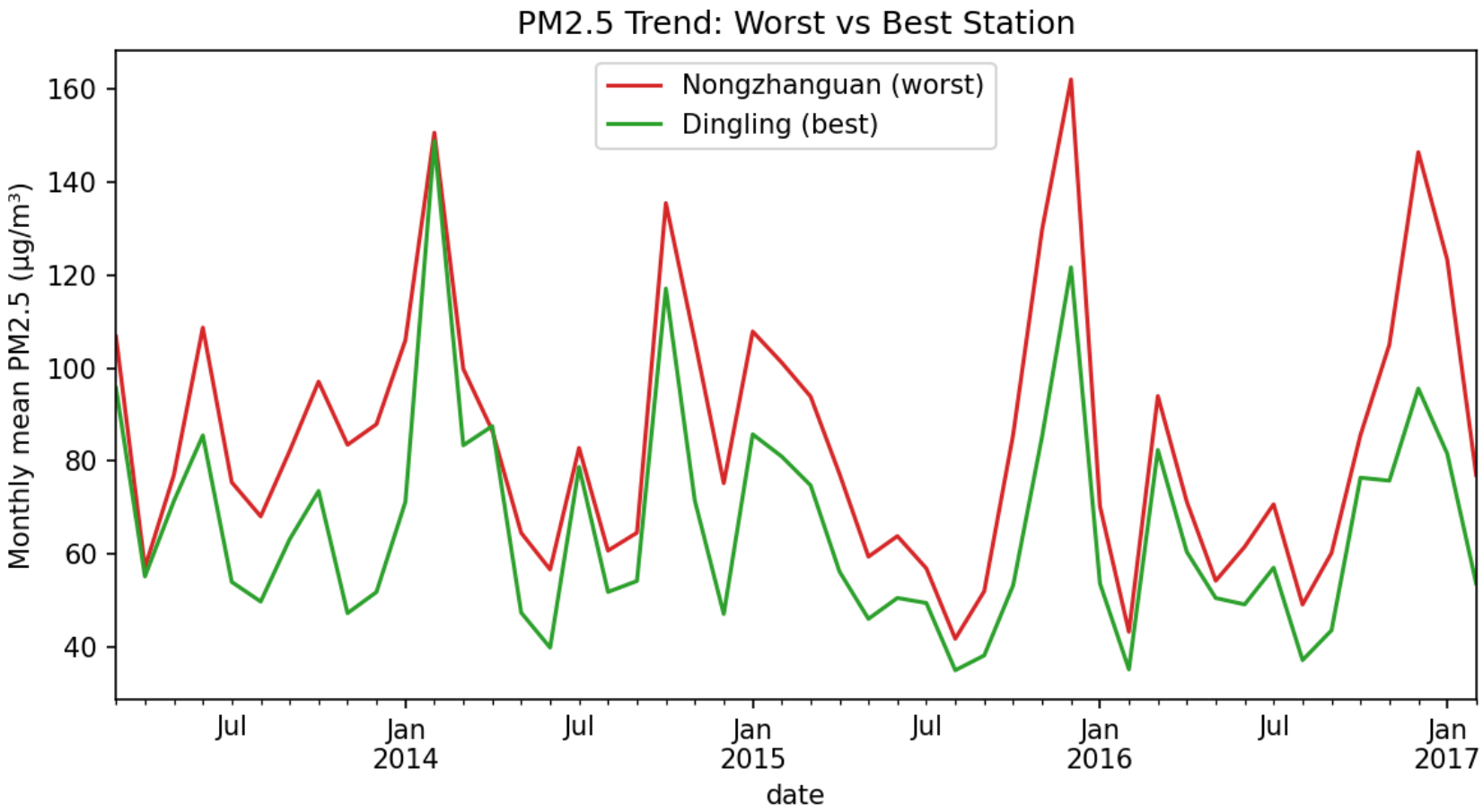
# Trend & Takeaways




**Findings**

- Strong winter peaks (heating season) at every site.
- Gap between worst and best widens in winter.
- Mild downward trend after 2014 — policy effect.

**Recommendation**

- Prioritise emission controls around Nongzhanguan, Dongsi and Wanshouxigong.
- Use Dingling as a clean-air baseline.